APPLIED MACHINE LEARNING & CLASSIFICATION

AIRLINE PASSENGER SATISFACTION PREDICTION USING MACHINE LEARNING

Analyzing multidimensional passenger flight experiences and predicting satisfaction states utilizing comparative supervised classification algorithms.

CANDIDATE DETAILS
[Student Name / Candidate]
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PROGRAMME & DEGREE
B.Tech in Computer Science & Eng.
Academic Capstone Project

FACULTY EVALUATION
Project Review Committee
[Department / University Name]

CONTEXT & MOTIVATION

Project Introduction & Industry Relevance

Problem Definition

What is Passenger Satisfaction Prediction?

It is a supervised machine learning task that models individual passenger feedback across flight phases (booking, check-in, in-flight entertainment, Wi-Fi, delays) to classify whether a passenger leaves **satisfied** or **neutral / dissatisfied**.

The Core Industry Problem

Commercial aviation is highly competitive with razor-thin margins. Traditional post-flight surveys suffer from low response rates, non-response bias, and delayed aggregation, preventing carriers from taking proactive operational remedies.

Why Machine Learning is Essential

ML models capture non-linear relationships and high-order feature interactions (e.g., how an arrival delay impacts a business-class traveler differently than a personal traveler), generating immediate, actionable predictive insights.

REAL-WORLD OPERATIONAL SCENARIO

The Flight Delay Dilemma

Consider a loyal frequent flyer in *Business Class* experiencing a 45-minute departure delay. If cabin cleanliness and inflight Wi-Fi are rated 5/5, the passenger is likely to stay satisfied. However, in *Eco Plus*, poor legroom coupled with the same delay flips the passenger into dissatisfaction.

> **Key Insight:** Machine learning pinpoints which services serve as effective buffers against unavoidable logistical disruptions like weather and air-traffic delays.

Significance to Aviation Stakeholders

- **Customer Retention:** Retaining existing high-tier travelers costs significantly less than acquiring new passengers.
- **Targeted Resource Allocation:** Guides ground and cabin operations to allocate budgets toward the highest-impact service touchpoints.
- **Data-Driven NPS:** Enables automated classification of satisfaction without manual auditing.

RESEARCH FRAMEWORK

Project Aim & Five Core Objectives

Academic Scope

PRIMARY PROJECT AIM

Analyze airline passenger experiences and predict satisfaction states using Machine Learning classification models.

Target: satisfaction (Binary)

01

Overall Satisfaction

Compute benchmark distribution and percentage shares across target categories to detect class balance vs. skew.

```
value_counts(normalize=True)
```

02

Travel Class Impact

Examine satisfaction proportions across Business, Eco, and Eco Plus cabins via crosstabs to evaluate tier expectations.

```
crosstab(Class, satisfaction)
```

03

Type of Travel

Evaluate behavioural divergence between passengers traveling on Business Travel versus Personal Travel itineraries.

```
crosstab(Type of Travel, ...)
```

04

Service Ratings

Isolate mean metric scores for 14 operational features (Wi-Fi, online boarding, legroom, food) by satisfaction status.

```
groupby("satisfaction")[services].mean()
```

05

Flight Delays

Quantify departure and arrival delay minutes in relation to final passenger satisfaction outcomes.

```
groupby("satisfaction")[[Delays]].mean()
```

DATA ARCHITECTURE

Dataset Attributes & Feature Taxonomy

Source: Airline Passenger Satisfaction CSV

1. Passenger Demographics

Features capturing identity: Age (Numerical), Gender (Male / Female), Customer Type (Loyal Customer / Disloyal Customer).

2. Flight & Itinerary Characteristics

Operational trip variables: Type of Travel (Business / Personal), Class (Business, Eco, Eco Plus), Flight Distance (Nautical/Statute miles).

3. Punctuality & Delay Dynamics

Time discrepancies: Departure Delay in Minutes and Arrival Delay in Minutes.

Target Variable: satisfaction

Supervised binary outcome: satisfied vs. neutral or dissatisfied (Encoded numerically as 1 and 0 for machine learning).

14 Inflight & Ground Service Rating Features

Scale: 0 to 5

Subjective operational ratings evaluated by passengers during post-trip surveys:

- Inflight wifi service
- Inflight entertainment
- Departure/Arrival convenient
- On-board service
- Ease of Online booking
- Leg room service
- Gate location
- Baggage handling
- Food and drink
- Checkin service
- Online boarding
- Inflight service
- Seat comfort
- Cleanliness

Design Note: Metadata columns (Unnamed: 0 and id) are intentionally pruned to eliminate synthetic variance.

ETL PIPELINE

Data Cleaning, Statistical Audit & Preprocessing

Pipeline Execution Steps

1. Pruning & Text Normalization

drop & strip

Dropped non-informative index columns (`Unnamed: 0`, `id`). Applied string stripping across all object columns to prevent whitespace mismatch.

2. Deduplication & Imputation

mean & mode

Removed duplicate passenger entries. Imputed missing numerical values via **mean** and categorical missing values via statistical **mode**.

3. Distribution Auditing (Skew & Kurtosis)

cov & corr

Computed correlation and covariance matrices alongside skewness and kurtosis metrics to assess deviation from standard normal distribution.

4. Outlier Auditing (IQR Method)

Q1 - 1.5·IQR

Identified extreme observations across Age, `Flight Distance`, `Departure Delay`, and `Arrival Delay` via Interquartile Range thresholds.

5. Feature Engineering: Age Group

pd.cut

Discretized continuous passenger ages into demographic lifecycle bins: `18-24`, `25-39`, `40-59`, and `60+`.

6. One-Hot & Label Encoding

get_dummies

Applied `pd.get_dummies(drop_first=True)` to convert categorical regressors into binary columns. Applied `LabelEncoder` to transform the target variable.

7. Stratified Train-Test Splitting

80:20 Split

Partitioned data with `test_size=0.20` using `random_state=42` and `stratify=y` to preserve class representation in both sets.

8. Feature Standardization

StandardScaler

Fitted `StandardScaler` on training features and transformed test features to prevent scale dominance in distance/gradient models (Logistic Regression & SVM).

PATTERN RECOGNITION

Exploratory Data Analysis & Visual Analytics

Analytical Visualizations

Satisfaction & Age Distribution

Target Countplot: Categorized counts of *satisfied* versus *neutral/dissatisfied*.

Age Histogram + KDE: Plotted across 30 bins to reveal age density curves and peak passenger cohorts (middle-aged travelers).

Visualizes baseline demographic spread.

Class & Travel Segmentations

Hue Countplots: Grouped evaluations tracking:

- **Class:** Business cabin vs. Eco / Eco Plus.
- **Type of Travel:** Business Travel vs. Personal Travel.
- **Customer Type:** Loyal vs. Disloyal passengers.

Identifies high-satisfaction segments.

Service Ratings & Delay Dynamics

Service Bar Chart: Mean comparison of 14 service features grouped by satisfaction.

Correlation Heatmap: Evaluates inter-feature collinearity and delay correlations.

Symmetric Log Boxplot: Arrival delay minutes scaled to inspect heavy-tail delay outliers.

Reveals operational service bottlenecks.

EDA Takeaway: Business class travelers demonstrate markedly higher satisfaction rates, while inflight Wi-Fi, online boarding, and legroom show the highest positive discrepancy between satisfied and dissatisfied passengers.

MODEL FORMULATION

Four Supervised Machine Learning Classifiers

Algorithm Comparison

1. Logistic Regression

LINEAR BASELINE

Working Principle: Maps linear combinations of weighted inputs through the Sigmoid function to output class probabilities between 0 and 1.

Why Used: Fast, computationally efficient benchmark for binary classification; highly interpretable coefficients.

Config: `max_iter=1000`, trained on `StandardScaler` features.

2. Decision Tree

NON-LINEAR PARTITIONING

Working Principle: Hierarchically splits feature space using purity metrics (Gini / Entropy) to form transparent if-else decision rules.

Why Used: Captures non-linear thresholds and interactions directly without requiring feature standardization.

Config: `max_depth=15`, `random_state=42`.

3. Random Forest

ENSEMBLE BAGGING

Working Principle: Constructs an ensemble of decorrelated decision trees using bootstrap aggregating and random feature subsets.

Why Used: Significantly suppresses variance, resists overfitting, and delivers superior generalization accuracy.

Config: `n_estimators=100`, `n_jobs=-1`.

4. Support Vector Machine

MAXIMAL MARGIN

Working Principle: Solves convex optimization to establish an optimal separating hyperplane that maximizes the geometric margin between classes.

Why Used: Robust against noise in high-dimensional feature spaces with rigorous theoretical foundations.

Config: `kernel="linear"`, sampled on scaled data (`n=15,000`).

PERFORMANCE ASSESSMENT

Model Training, Validation & Evaluation Protocol

Evaluation Protocol

Validation Methodology

- **Data Partitioning:** 80% Training partition for parameter estimation; 20% holdout Testing partition for unbiased inference.
- **Stratification:** Target proportion preserved across sets to maintain identical ground-truth class distribution.
- **Selection Criteria:** The final champion model is strictly determined based on highest **Weighted F1 Score**.

Diagnostic Metrics Computed

Accuracy: Ratio of total correct predictions.

Precision: Exactness of positive classifications.

Recall: Sensitivity in catching true satisfaction.

Weighted F1: Harmonic mean weighted by support.

Evaluation Artifacts Generated in Code

The verification pipeline executes four systematic diagnostic checks for each classifier:

Evaluator Artifact	Role in Faculty Evaluation & Viva
Confusion Matrix	Exposes True Positives, False Positives, False Negatives, and True Negatives.
Classification Report	Per-class breakdown of Precision, Recall, F1, and class support counts.
Tree Visualization	Visual audit of top 3 decision tree levels for structural interpretability.
Sorted Summary Table	Ranks all 4 algorithms to identify the best-fit classifier automatically.

Academic Integrity Principle: Accuracy figures and rankings are computed dynamically via code execution without hardcoded values.

END-TO-END SYSTEM

System Architecture, Practical Applications & Trade-Offs

Implementation Flow

COMPLETE PIPELINE FLOW

Raw CSV → Data Cleaning → Preprocessing → EDA & Audit → Feature Eng. → Train-Test Split → 4 ML Models → Evaluation → Best Model

Commercial Aviation Applications

- **Service Touchpoint Enhancement:** Isolates underperforming amenities (e.g., poor in-flight Wi-Fi or slow boarding) for capital upgrading.
- **Customer Experience Management (CXM):** Directs customer success teams to proactively resolve feedback from high-value business flyers.
- **Delay Recovery Strategies:** Identifies compensatory in-cabin services that can mitigate frustration caused by flight delays.
- **Executive Decision Support:** Provides quantitative data backing fleet refurbishment and ground staffing allocations.

Core Advantages

End-to-end reproducible pipeline, rigorous data imputation, multi-model cross-comparison, and objective selection via Weighted F1 metric.

Operational Limitations

SVM requires downsampling for practical training times; survey responses remain inherently subjective across individual passenger expectations.

PROJECT SUMMARY

Conclusion, Future Enhancements & Q&A

Project Wrap-Up

Academic & Technical Achievements

- Successfully formulated, cleaned, and explored a comprehensive airline satisfaction dataset.
- Developed and bench-marked four core classification paradigms (Logistic Regression, Decision Tree, Random Forest, SVM).
- Demonstrated that systematic machine learning models provide an objective, data-driven approach to evaluating passenger experience.
- Proved that ensemble methods reduce variance when analyzing diverse customer feedback patterns.

Roadmap for Future Research

- **Web API Deployment:** Package the best classifier into a lightweight Flask / FastAPI backend for real-time inference.
- **Continuous Model Validation:** Test generalization against newly collected flight datasets from diverse international routes.
- **Advanced Feature Enrichment:** Incorporate sentiment analysis from open-ended passenger text comments.
- **Model Monitoring:** Implement drift detection to monitor seasonal variations in passenger ratings.

THANK YOU

Faculty Evaluation & Viva Voce Defense Session

Questions & Discussion